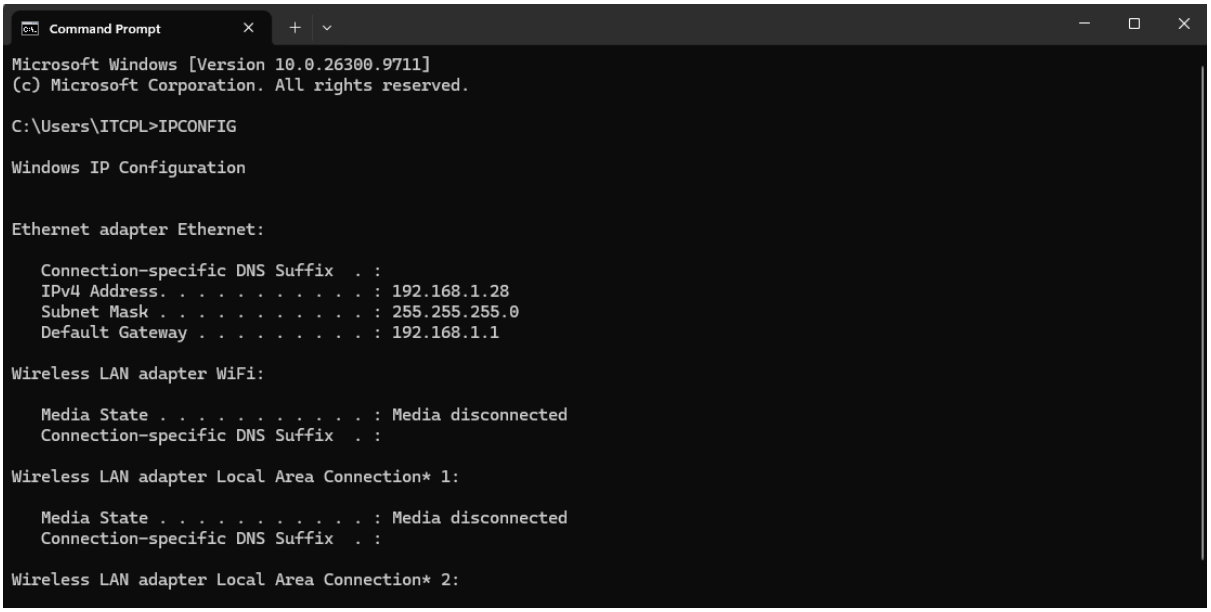
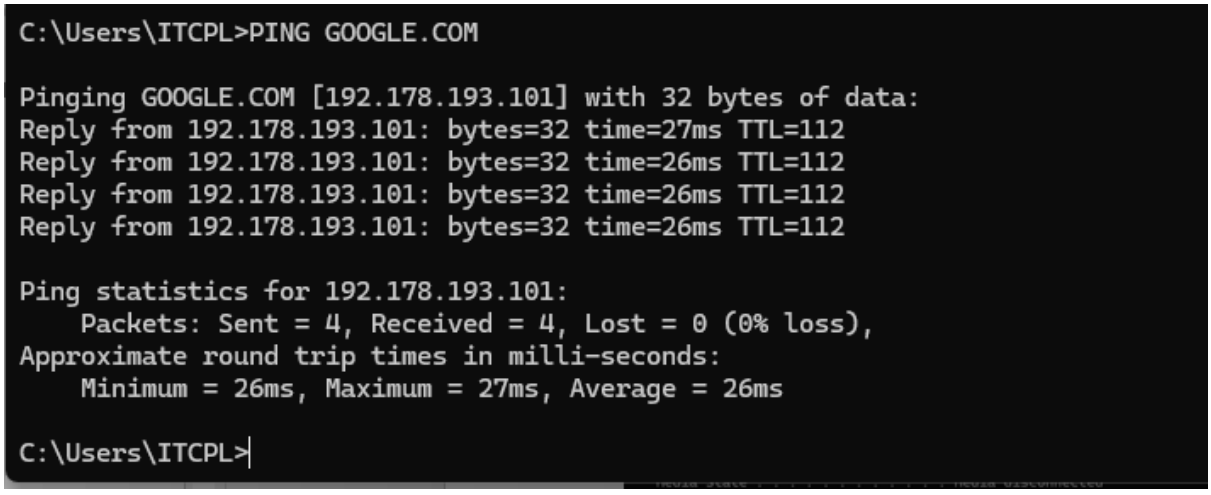
1) Open the command prompt and use the ipconfig tool to find your computer's IPv4 address and default gateway, then take a screenshot of the output.



2) Use the ping command to check the connectivity between your computer and www.google.com, then note down the average response time shown.



3) Run the tracert (or traceroute on Linux/Mac) command to www.flipkart.com and list the number of hops it takes to reach the destination.

```
C:\Users\ITCPL>.emulator_console_auth_token

C:\Users\ITCPL>tracert www.flipkart.com

Tracing route to e348826.dsca.akamaiedge.net [23.55.244.178]
over a maximum of 30 hops:

  1     1 ms    <1 ms    <1 ms  gpon.net [192.168.1.1]
  2     2 ms     2 ms     2 ms  10.230.192.1 [10.230.192.1]
  3     3 ms     2 ms     2 ms  10.111.24.2 [10.111.24.2]
  4     *        *        *     Request timed out.
  5    13 ms    13 ms    13 ms  103.241.47.110
  6    13 ms    13 ms    13 ms  49.44.129.94
  7    11 ms    10 ms    11 ms  a23-55-244-178.deploy.static.akamaitechnologies.com [23.55.244.178]

Trace complete.

C:\Users\ITCPL>
```

4) Use the nslookup tool to find the IP address of www.spotify.com and explain in one line what this tool helps you verify.

```
C:\Users\ITCPL>nslookup www.spotify.com
Server:  gpon.net
Address:  192.168.1.1

Non-authoritative answer:
Name:    atc.spotify.map.fastly.net
Addresses:  2a04:4e42:19::810
          151.101.67.42
          151.101.195.42
          151.101.131.42
          151.101.3.42
Aliases:  www.spotify.com


C:\Users\ITCPL>
```

**5) A friend says they cannot access Zomato's website from their laptop. Suggest a sequence of three troubleshooting commands (choose from: ping, ipconfig, tracert, nslookup) you would run to diagnose the issue, and briefly state what each command would help you check.

Hint: Think about checking both network connectivity and DNS resolution.**

1. ipconfig – Checks the laptop's IP address, subnet mask, and default gateway to verify network configuration.

2. ping 8.8.8.8 – Checks whether the laptop can reach an external IP address and confirms basic network connectivity.

3. nslookup www.zomato.com – Checks whether DNS can resolve Zomato's domain name into an IP address.

Conclusion: These three commands help identify whether the issue is caused by incorrect network settings, a connectivity problem, or DNS resolution failure.